

# Victor Gusev

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## Education

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**University of California, Los Angeles (UCLA)**

Expected June 2028

B.S. in Computer Science and Engineering

GPA: 3.64/4.00

*Coursework:* Data Structures & Algorithms, Object-Oriented Design, Probability, Logic Design

## Technical Skills

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**Languages:** Python, C++, Java, C, JavaScript

**Frameworks/Libraries:** PyTorch, React, Node.js, Express, Socket.io, ROS, NVIDIA Isaac Lab/Sim

**Technologies:** MongoDB, Docker, Linux, Git, LiDAR, reinforcement learning (PPO), PID control, RViz

## Experience

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**Software Engineering Intern** | [Dimaag.ai](#)

June 2026 – August 2026

- Developed and deployed a hybrid reinforcement-learning and classical local planner for autonomous-vehicle trajectory tracking, reducing cross-track error by 80% versus the classical planner.
- Deployed and validated navigation policies on a physical autonomous vehicle, maintaining <10 cm mean trajectory error at speeds exceeding 10 mph on complex paths.
- Built PPO training and 2D LiDAR simulation pipelines in NVIDIA Isaac Lab and Gazebo, using domain randomization and cross-simulator testing to transfer policies from simulation to hardware.
- Developed and tested Python modules within a shared robotics codebase using Git-based code review and containerized development workflows.

## Projects

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**Proof** | React, MongoDB, Web Scraping, Gemini API | [Website](#)

- Built and deployed a React/MongoDB PWA for recipe versioning and ingredient inventory, with 100+ automated tests covering core application workflows.
- Built an ingestion pipeline combining web scraping and Gemini API extraction to convert recipes from arbitrary URLs into structured application data.
- Implemented synchronized cross-device timers with Socket.io for real-time tracking of multi-stage cooking workflows

**Micromouse Simulator** | Java, Flood-Fill Pathfinding, Simulation | [GitHub](#)

- Built a Java Micromouse simulator implementing flood-fill pathfinding that recomputed routes as maze walls were incrementally discovered.
- Implemented frame-by-frame visualization and replay tooling for debugging routing decisions on real competition mazes.

## Leadership

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**Robotics Programming Lead** | [FIRST Robotics](#) | Team 1458

August 2021 – May 2025

- Led software development for a 40+ member FIRST Robotics team, implementing PID control, vision tracking, LiDAR localization, and autonomous routines for competition robots.
- Developed autonomous and vision systems for a team that qualified for the FIRST World Championship and ranked in the top 5% regionally.